MILITARY SPECIFICATION

CRANE, WHEEL MOUNTED, HYDRAULIC, LIGHT, 7-1/2 TON

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is availailable for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-propelled, diesel-powered, pneumatic tired, full revolving, hydraulically operated 7-1/2 ton cranes.

1.2 Classification. Cranes covered by this specification shall be of the following types, as specified (see 6.2):

Type I - General Purpose
Type II - Airborne/Airmobile

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-W-410 - Wire Rope and Strand.

Beneficial comments (recommendations, additions, deletions) and any-pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir. VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MILITARY

MIL-V-173	Varnish, Moisture-and Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	Treatment and Painting of Materiel.
MIL-L-2104	Lubricating Oil, Internal-Combustion Engine, Heavy-Duty.
MIL-L-2105	Lubricating Oil, Gear, Multipurpose.
MIL-C-3580	Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full-Revolving, and their Attachments, Packaging of.
MIL-G-3859	Grease Gun, Hand-Operated, Lever, Push and Screw Type.
MIL-M-7866	Molybdenum Disulfide, Technical, Lubrication Grade.
MIL-A-8421	Air Transportability Requirements, General Specification for.
MIL-G-10924	Grease, Automotive and Artillery.
MIL-A-46153	Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-B-46176	Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative.
MIL-D-51246	Decontaminating and Reimpregnating Kit, Individual, M13.
MIL-H-52471	Hose and Hose Assemblies, Rubber, Hydraulic Pressure Type, General Specification For.

STANDARDS

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.
MIL-STD-669	- Loading Environment and Related Requirements for Platform Rigged Airdrop Materiel.
MIL-STD-814	- Requirement for Tiedown, Suspension and Extraction Provisions on Military Material for Airdrop.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MS24207	- Coupling, Greased Gun, Hydraulic Type Nozzle.

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MS35000	Battery, Storage, Lead-Acid, Waterproof.
MS51318	Headlight Blackout, 24 Volt, Waterproof.
MS51330	Stoplight, Taillight, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.
MS51118	Pintle Assembly, Towing -40$000 Lbs Capacity, Manual Release.
MS52131	Connector, Plug, Electrical Inter Vehicle power Cable.
MS53052	Bracket Assembly, liquid Container, Five Gallon.
MS500004	Pad Eye, Vehicular, Towing Lug.
MS500048	Towbar, Motor Vehicle: Light, Medium and Heavy Duty.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, form a part of this specification to the extent specified herein.

ENVIRONMENTAL PROJECTION AGENCY

Protection of Environment
Title 40 Code of Federal RegulationS (CFR)

DEPARTMENT OF AGRICULTURE

Forest Service Standard -5100-1
Standard for Spark Arrestors for Internal Combustion Engines

DEPARTMENT OF LABOR

Occupational Safety and Health Administration (OSHA) Standards.
Title 29 Code of Federal Regulations (CFR), Part 1926.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 other publications. The following document(s) forma part of this specification to the extent specified herein. Unless otherwise indicated; the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY, INC (AWS)

D14.3 - Specification for Earthmoving and Construction Equipment Welding.

(Application for copies should readdressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

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POWER CRANE AND SHOVEL ASSOCIATION

Mobile Hydraulic Crane Standards (PCSA Standard No. 2)

(Application for copies should be addressed to the Power Crane and Shovel Association, Marine Plaza, Suite 1700, 111 E. Wisconsin Ave., Milwaukee, WI 53202.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B30.15 - Safety Standards for Mobile Hydraulic Cranes.
Z53.1 - Safety Color Code for Marking 'Physical Hazards.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

ASSOCIATION OF AMERICAN RAILROADS

Rules Governing the Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.,

Agent, National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street NW, Washington, D.C. 20036.)

TIRE AND RIM ASSOCIATION INC.

Tire and Rim Association Yearbook.

(Application for copies should readdressed to the Tire and Rim Association Inc., 3200 West Market Street, Akron, Ohio 44313.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago,Illinois 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The crane shall consist essentially of a 2- and 4-wheel drive, 2- and 4-wheel steer, pneumatic-tired, diesel-engine powered chassis and a center mounted, hydraulically operated, full revolving telescoping boom. The crane shall be either type I or type II as Specified (See 6.2). The type I is intended for general purpose use. The type II is intended for use by airborne and airmobile units. The type II crane shall be identical to the type I crane except for changes required to meet additional transportability requirements for the type II crane. As required by law for commercial sales, the crane shall meet all applicable requirements of the Occupational Safety and Health Administration (OSHA) Standards and the Environmental Protection Agency (EPA) Emission Standards ineffect on the date of the opening of bids. Also, the crane shall meet all requirements set forth for compliance with the Power Crane and Shovel Association Standard No. 2 and American National Standards Institute B30.15. The crane shall be based on a current commercial production model which has demonstrated industry acceptability by having been manufactured, marketed, and sold to commercial users for at least one year prior to the issuance of the request for proposal or information for bid, or a model that has been produced and successfully furnished to the Government within three years of the issuance of the request for proposal or information for bid.

3.1.1 Weights and dimensions. The overall weight of the type I and type II crane shall not exceed 26,000 pounds and the axle loads specified in 3.7.1 shall not be exceeded. The overall height of the type I crane shall not exceed 102 inches with minor disassembly. Any disassembly and reassembly required shall be accomplished by one person using only common hand tools and special tools furnished with the crane. The overall height of the type II crane shall not exceed 96 inches and when rigged on air delivery platform for airdrop (see 3.7.2.1) the type II crane shall be shaped to fit under the aircraft tipoff curve using figure 1 and figure 9 in MIL-STD-814. The overall width of the type I and type II crane, excluding normal tire bulges shall not exeed 96 inches. The overall length of the type I and type II crane shall not exceed 27 feet. The length of the carrier of the type I and type II crane shall not exceed 20 feet.

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3.1.2 Performance. The crane shall be capable of:

a. Lifting rated loads (see 6.8.1) of not less than 7-1/2 tons at a 10-foot operating radius and 2 tons at a 28-foot operating radius (see 6.8.2) through 360 degrees with outriggers extended and lifting rated loads of not less than 7-1/2 tons at a 10-foot operating radius over the front with the carrier stationary and outriggers retracted.
b. Negotiating 30 percent longitudinal and 15 percent side slopes of firm soil without hook load.
c. Unassisted starting and operation in any ambient temperature between minus 25° F and plus 120° F and at any altitude between sea level and 5000 feet.
d. Operating safely in ammunition and petroleum, oil and lubricant (POL) supply points.
e. Fresh water fording to a depth of not less than 30 inches without damage or leaking of water into lubricant reservoirs.
f. Attaining a travel speed of not less than 20 miles per hour on firm level ground or pavement; and travel cross country on unimproved, off-road terrain at maximum safe speeds.
g. Lifting and carrying suspended palletized loads (40 inches long X 48 inches wide X 62 inches high) of not less than 4000 pounds at load radius of not less than 20 feet over primary, secondary, and unimproved roads at speeds up to 2-1/2 miles per hour.
h. A curb to curb turning diameter of 40 feet or less in diameter in accordance with SAE J695 utilizing four wheel steer.
i. A swing speed of not less than 1.75 rpm and not more than 3.5 rpm at governed engine speed with the swing control in the full open position.

3.2 First article (preproduction model). The contractor shall furnish one or more of each type I and type II cranes as specified (see 6.2) for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce cranes that comply with the requirements of this specification. Examinations and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.2). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests, as specified. For the type II crane, following completion of the examination and tests by the contractor, the Government shall perform a simulated airdrop test at a site selected by the Government. Acceptance of the preproduction model(s) of the type II crane will be dependent on successful completion of these tests as well as all contractor conducted examinations and tests. Any changes or deviations of production cranes from the approved preproduction model(s) during production will be subject to the approval of the contracting officer. Approval of the preproduction model(s) will not relieve the contractor of his obligation to furnish cranes conforming to this specification.

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more production cranes for examination and tests as specified in 4.4.

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3.4 Material. All material used in the manufacture of the crane shall be of the same quality as that being used by the contractor in his comparable production cranes for commercial market sales. Material shall be free from any defects or imperfections, that would affect the crane's operations or serviceability (see 6.9). All material shall be inherently corrosion resistant or suitably protected against corrosion and deterioration from the environment as specified in 3.14 and 3.22.

3.4.1 Keys and key slots. Keys and key slots shall conform to SAE J502 and J503.

3.4.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable.

3.5 Safety, human factor, and sound level.

3.5.1 Safety. All rotating, reciprocating, or moving parts shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard, All hot surfaces of the equipment, including exhaust pipes and other lines which maybe subject to high temperatures or exposed to contact by personnel, shall be fully guarded or insulated. The engine exhaust system discharge shall be so directed that it does not endanger personnel. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. The contractor shall apply the Government furnished Electrical Warning Decals (see 3.24) according to the instruction provided by the contracting officer. Portions of the superstructure that extend out and over the carrier when being swung where it may cause a strike or crushing hazard for personnel shall have the lower and back area taped in color coding in accordance with ANSI 253.1.

3.5.2 Human factors. The operating and maintenance characteristics of the crane shall provide for operating and maintenance personnel ranging from the 5th percentile female, through the 95th percentile male, arctic clothed, in acordance with ME J833 and SAE J925. The size and location of access/egress passages shall be in accordance with SAE J185. Components and equipment removed, disassembled or relocated for operation or transportability as specified in 3.1.1 and 3.7, which must be lifted or carried by operating personnel shall not exceed 50 pounds in weight for one person lifts and 100 pounds for two or more person lifts and shall be provided with suitable handles or grasp areas.

3.5.3 Sound level. The crane shall be equipped with all available standard and optional sound suppression equipment. If the noise level in the area occupied by the operator and personnel normally working with the crane is 85 dB(A) or greater when measured in accordance with SAE J919, caution plates shall be permanently affixed to the crane in conspicuous protected locations in the operator^{s} compartment and where other personnel normally work. The caution plate shall state: “Caution: Hearing protection required." The plate shall have a yellow background with black lettering and be made of corrosion resistant material. If the sound level with cab doors or windows open is 85 dB(A) or greater when it is less than 85 dB(A) with the doors or windows closed, a caution plate stating the following shall be provided: "Caution: Hearing

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protection required while operating with doors or windows open." The exterior sound level of the crane shall not exceed 88 dB(A) when measured in accordance with SAE J88. Manuals shall contain appropriate discussions concerning noise hazards and hearing protection.

3.5.4 Back-up alarm. A back-up alarm shall be provided on the crane and shall comply with SAE J994, type C. The back-up alarm signal shall be audible above and distinguishable from the surrounding noise level. The alarm shall be furnished with means for deactivation during tactical conditions.

3.5.5 Horn. The crane shall be equipped with a hand or foot operated horn which is dinguishable from and not less than 10 dB(A) above the crane noise level when tested in accordance with 4.6.2.21.2.

3.6 Maintenance.

3.6.1 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility.

3.6.2 Filling, draining and checking previsions. The hydraulic system reservoir, the transmission housing, and all other enclosures which contain lubricant or hydraulic oil shall be equipped with dipsticks, sight gages, or with checkplugs conforming to SAE J371 to determine the fluid levels. Each enclosure shall be equipped for filling and for draining. Each drainplug or valve shall be located so that removal of the plug or valve will result in complete drainage of the fluid from the enclosure (except torque converter). When the crane is in a level position, drainage of fluids shall be to a container below the crane without draining on or over any part of the crane. Integral tubes, hose, or troughs may be used to convey the fluids from the drain outlet. Accessibility to the fill opening, the fluid-level checking device, and the drainplug or valve shall be provided without removal or adjustment of accessories or parts, except for plates equipped with hand-operable, quick disconnect fasteners. The radiator drain shall be such that coolant will not drain onto any part of the crane. Extension hose is permitted.

3.6.3 Oil sampling valve. An oil sampling valve shall be provided on the engine and transmission. The valve shall be manually operated and shall close automatically after release. It shall be made of material resistant to corrosion such-that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way that samples can be taken easily and without being exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken it shall be a true representation of oil that is flowing when the engine is running.

3.7 Transportability. The crane shall be capable of being transported worldwide by aircraft, highway, rail, and marine modes and withstand the impact forces encountered without damage or deformation.

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3.7.1 Type I and type II crane. The crane shall be capable of being transported on a railroad car, in accordance with Association of American Rail-Roads “Rules Governing the Loading of Commodities on Open-Top-Cars" and shall not exceed the PPI clearance dimensions shown in figure 2. The crane shall be capable of air transport by C-130, C-141, and C-5 aircraft in accordance with MIL-A-8421 and the axle loads shall not exceed 13,000 pounds per axle. Disassembly prior to and reassembly after transporting shall be accomplished in not more than one hour using only common hand tools and special tools furnished with the crane. Cutting and welding are not permitted. Removal of components shall not prevent the crane from self-loading under its own power.

3.7.2 Type II crane.

3.7.2.1 Parachute (airborne). The type II crane shall be capable of being loaded on an air delivery platform 24 feet long and transported by both C-130 and C-141 aircraft. It shall be capable of being airdropped from a C-130 by both low-velocity airdrop and Low Altitude Parachute Extraction System (LAPES) airdrop and from a C-141 by lo-velocity airdrop only. The platform rigging shall be in accordance with MIL-STD-669 with the tires installed on the crane. The following shall also be met:

a. The overall height of the rigged load shall not exceed 100 inches and the crane shall be shaped to fit under the aircraft tipoff curve as determined by the use of figure 1 and figure 9 in MIL-STD-814.
b. There shall be not less than 12 inch thickness of honeycomb material under any portion of the underside of the crane except for the tires with not less than 82.5 square feet of contact area for the honeycomb material. The need for load spreading lumber shall be held to a minimum. Open spaces shall be bordered by supports that are level and not more than 18 inches apart and capable of supporting a force of 7,000 pounds times the area of honeycomb material in square feet that is supported by these supports.
c. The static force exerted by each tire against the platform shall not exceed 1,000 pounds. The tires shall be capable of withstanding the effects of the rim traveling through the 9 inch honeycomb crushing stroke on impact without damage or loss of inflation pressure.
d. The crane shall be capable of withstanding a short duration impact force of approximately 20 G's in accordance with MIL-STD-669 without breakage or permanent deformation of any structural member or component or degradation of the crane’s ability to perform its intended function.
e. Material or equipment may be disassembled or removed to comply with the height requirements of the rigged load provided that this material or equipment can be loaded on the same platform and can be reassembled or reinstalled by not more than two men in not more than 1 hour utilizing the inherent capability of the crane and tools furnished with the crane.
f. To aid in the rigging for parachute delivery no hydraulic, air, oil, fuel and electrical lines shall be installed in unprotected location on the underside of the carrier frame.

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3.7.2.2 Helicopter (airmobile). The type II crane, equipped with the reassembly kit (see 3.8) shall be capable of being transported externally by helicopter in not more than two loads. Each load shall not exceed 15,000 pounds. The time required to assemble the crane into an operational unit shall not exceed 2 hours after delivery of the two loads and shall be accomplished by not more than four persons utilizing only the material, tools, and equipment delivered in the two loads. Components or material may be removed from one load to reduce weight and delivered with the other load providing that all other requirements are met. The powered section shall be drivable and steerable under its own power for a distance of not less than 50 feet over level, flat, unimproved areas.

3.8 Reassembly kit. When specified (see 6.2), if required for the reassembly of the type II crane from two loads after delivery by helicopter, a reassembly kit shall be provided for the type II crane. The reassembly kit shall be part of the two loads, and the components of the kit shall be of such size, and shape that can easily be handled by no more than the four persons allowed to assemble the kit, reassemble the crane, and disassemble and stow the kit in less than 2 hours. The inherent capabilities of the crane, especially the hydraulic and electrical systems may be used to assist in the reassembly. The kit shall include caps, covers, plugs, guards, or other means of protecting disconnected hydraulic, air, lubricant, fuel and electrical lines, and exposed parts from any contamination or damage that would adversely affect the operation of the crane, These protective means shall be reusable and where practical shall be permanently attached to the fitting or part which it will protect for easy access and loss prevention. Any line disconnected in the sectionalization process shall incorporate quick disconnect fittings and shall be sized, keyed or otherwise be identified with its mate to facilitate reconnection. The kit shall also include all special tools required in the reassembly. Provisions shall be furnished for storing the kit on the crane, full operation of the crane with the kit stored on board, and disassembly of the crane using the kit for helicopter transport.

3.9 Carrier. The crane shall be provided with a chassis with necessary braces and reinforcements that will provide adequate mobility and support under all operating conditions. Two tow lugs, conforming to MS500004, 10-50 ton, to mate with towbar, MS500048 medium duty, shall be furnished and installed on the front of the carrier. A swivel type towing pintle conforming to MS51118 or of commercial design raced at not less than 40,000 pounds capacity shall be furnished on the rear of the carrier.

3.9.1 Engine. For the purpose of selecting the engine, the crane shall be classified class II end item in accordance with MIL-STD-1410. The engine, including all systems, components and accessories, shall conform to MIL-STD-1410, except as specified herein. The engine shall be equipped with a cold starting aid system of the glow plug or ether primer type, conforming to paragraphs 4.2.1 and 4.2.12.5 of MIL-STD-1410. The engine furnished shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. The engine shall meet all operational requirements and conditions specified herein without exceeding the engine manufacturer's recommended operating limits for this application.

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3.9.1.1 Fuel system. The fuel system shall be so located that no part of the system extends beyond the widest part of the crane and that no part of the fuel tank is forward of the front axle. The fuel tank joints shall be closed by welding, brazing, silver soldering, or equivalent technique. Joint closing by crimping or lead-based or soft soldering shall not be used. The capacity of the fuel tank shall be sufficient for not less than 200 miles of highway travel at maximum attainable speeds up to 20 mph. The fuel tank shall be legibly marked "DIESEL FUEL ONLY" in block letters 1/2 inch in height. The fill pipe opening shall be located outside of the operator's compartment and located such that no fuel spilled during fueling will contact any part of the exhaust or electrical systems. The tank filling components shall include a strainer. The fuel tank shall be provided with shut-off valves for all lines to the tank. Fuel lines and fittings shall be located and routed in protected locations not subject to chafing or damage. The mounting of the fuel tank shall prevent shiftings damage, and fatigue as a result of travel and transports The fuel tank and the fuel filter housing shall be equipped with means for draining water and sediment from the fuel system.

3.9.1.2 Engine air induction system. The air induction system shall include a precleaned or dust scavenger, a heavy-duty dry type air cleaner and an air cleaner restriction indicator. The air cleaner restriction indicator shall be mounted in a location clearly visible to the operator without requiring removal of panels, covers, or lids.

3.9.1.3 Charging system. An engine driven alternator charging system shall be furnished. The system shall maintain battery charge and supply electricity to operate the electrical components.

3.9.1.4 Engine housing. The engine housing shall provide for easy access to both sides of the engine.

3.9.1.5 Exhaust system. The exhaust system shall be protected against entry of rain. The exhaust system shall be furnished with a spark arrestor conforming to the Forest Service Standard 5100-1 unless it is certified by the engine manufacturer as not being required for operations in environments found at ammunition, petroleum, oil and lubricant supply points.

3.9.1.6 Starter interlock system. Means shall be provided which prevent the engine from being started unless the transmission is in park or neutral or the clutch is disengaged.

3.9.1.7 Tachometer drive. The engine shall be furnished with a heavy duty key drive tachometer take-off in accordance with SAE J678. The drive shall be 5/32-inch size and shall have an outer threaded male connection with 7/8-18Ns-2A threads. The take-off shall be furnished with a threaded captive chained cap.

3.9.2 Transmission system. A full power shift transmission with torque converter shall be provided. The transmission system shall provide adequate gearing for all performance conditions specified herein and operational any speed up to maximum speed. The crane shall be capable of accelerating from a

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standstill to maximum speed under continuous full engine throttle on level ground and shall be accomplished without damaging the transmission The transmission shall be protected from damage when the operator moves or attempts to move the directional control to the opposite direction while the crane is in motion. All operating controls shall be operable from the operator's compartment. The system shall provide for both 2-wheel and 4-wheel drive at the selection of the operator. All transmission component material shall be compatible with operation with oil conforming to MIL-L-2104, grade 10 and MIL-L-46167. The transmission system shall be furnished with filter(s) having replaceable elements and sized in accordance with the transmission manufacturer's recommendation.

3.9.3 Front and rear axles. The crane shall be furnished with front and rear axles of the steering and driving type. The axle manufacturer's rating for torque input, gearing and beam loading shall be equal to or exceed the actual maximum loads imposed under all operating conditions. Axle installation shall provide for axle oscillation to provide adequate off road (see 6.8.8) capability and the means for restricting the oscilliation when lifting rated loads on tires and when being transported by helicopter for the type II crane.

3.9.3.1 Steering system. A hydraulic or a mechanical power assisted steering system shall be furnished to provide full control of the crane under all operating conditions. The steering system shall provide for front-wheel steering when the engine is dead. Front-wheel steering shall be controlled by a steering wheel. Rear-wheel steering shall be controlled by the same steering wheel or a tiller. An indicating light shall be provided on the instrument panel to indicate when the gearwheels are positioned other than straight ahead. The steering system shall provide the operator with a choice of front-wheel, or a combination of front- and rear-wheel steering, including crab steer capabilities. If a full hydraulic steering system is furnished on both axles, the crane shall be capable of negotiating the test courses in SAE J53 at 5 mph and 10 mph and the required steering wheel effort to negotiate the course shall not exceed limits specifed in SAE J53.

3.9.3.2 Tires and rims. All tires and rims shall be identical. Tires shall be of the tubeless type. The rated load carrying capacities shall be in accordance with the Tire and Rim Association Yearbook recommendations, Tires shall have traction-lug type tread and shall be sized to provide for not less than 14 inches ground clearance below the lowest part of the crane. The tires used shall be approved for the application by the tire manufacturer.

3.9.3.3 Fenders. The crane shall be equipped with fenders at all four wheels. The fenders shall be of adequate size and strength to allow their use as boarding and servicing provisions as specified in 3.9.6. The fenders may be removed and set aside to meet the weight restrictions in 3.1.1, 3.7.2.1, and 3.7.2.2 provided that they are installed with SAE standard nuts and bolts.

3.9.4 Outriggers. Four independently operated (hydraulic) outriggers capable of extension to not less than 6 inches below ground level, shall be provided. The outriggers shall be operated from the operator's compartment and shall not collapse due to loss of hydraulic pressure in the supply lines. The

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outriggers pads shall be attached to the outriggers beams by swivel joints which permit displacement of the pads by not less than 3 inches between opposite sides on opposite corners of the pad without breakage, damage or deformation. An indicator shall be provided for the operator to determine levelness of the crane during operation of the outriggers.

3.9.5 Operator's compartment. All controls for the carrier and crane shall be located in the operator's compartment in accordance with SAE J898 and labeled in accordance with SAE J298. All controls not covered by SAE J298 shall be identified as to use or function on plates or decals conforming to 3.18. For the controls covered by SAE J297, locations and movements shall conform to SAE 3297. The operator's seat shall be in accordance with SAE J899 and shall be provided with a seat belt in accordance with SAE J386, type I. Glare reducing side mirrors shall be -furnished and installed on each side of the crane to provide the operator with-rear view visibility.

3.9.5.1 Instrument panel. An instrument panel conforming to SAE 3209 and labeled in accordance with ME J298 shall be furnished. Those gages, indicators and switches not covered by SAE J298 shall be identified by plates or decals conforming to 3.18. The instrument panel shall include not less than the following gages or indicators:

Engine oil pressure gage
Engine coolant temperature gage
Converter oil temperature gage
Air Pressure gage, when applicable
Battery generator indicator (Ammeter or volt meter)
Fuel gage
Tachometer
Hourmeter (may be engine mounted)
LOW air pressure warning light and buzzer, when applicable
Rear steer indicator warning light - red
Turn signal indicator light (may be steeriag wheel column mounted)
Emergency flasher indicator light (maybe steering wheel column mounted)
Hi - Lo beam indicator light

The following switches shall be provided:

Ignition
Heater and defroster
Windshield wiper
Driving lights
Floodlights
Turn signals (may be steering wheel column mounted)
Emergency flasher (maybe steering wheel column mounted)
Blackout lights
Horn button (may be steering wheel mounted)

All gages, indicators, and switches shall be provided with adequate illumination for the operator to read all labels and gage indications in the absence of any other lighting.

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3.9.5.2 Crane capacity chart. A crane capacity chart shall be provided in accordance with ANSI B30.15 and shall contain not less than the following information:

a. Rated loads, in pounds, in not more than 5-foot increments for the full range of operating radii with and without outriggers.
b. That the rated loads do not exceed 85 percent of the tipping load.
c. Definition of the load to be lifted.
d. Hook block reeving.
e. That the crane has been tested in accordance with SAE J1063 and SAE J765.
f. Rated loads that can be lifted and carried at safe radius and travel speed.

3.9.5.3 operator's cab. A removable cab conforming to ANSI B30.15 and PCSA Standard NO. 2 shall be provided for the operator's compartment. An electric operated windshield wiper with not less than two speeds shall be provided for the front window. The cab shall permit operation by a 95th percentile male wearing arctic clothing as defined in SAE J833 and J925. The cab may be removed and set aside on the type II crane in order to meet the height restrictions in 3.1.1 and 3.7.2.1 and the weight restrictions in 3.1.1- and 3.7.2.2 provided crane operations are not affected. If the boom is lowered below the cab height to meet the height restrictions, means shall be provided to prevent any interference when the cab is installed.

3.9.5.4 Heater and defroster. Unless otherwise specified (see 6.2), a heater with two or more fan speeds and defroster shall be furnished for the operator's compartment. The heater shall be capable within 1 hour of reaching and maintaining a uniform cab temperature (+9° F) of 41° F or higher at an ambient temperature of -25° F. The defroster may be either part of a heater-defroster or a separate defroster fan and shall provide defrosting of the front and side windows. If a fan type defroster is provided, it shall be safety guarded, adjustable, and provided with an on/off switch. If the heater is fuel fired, it shall conform to SAE J1024.

3.9.6 Boarding and servicing provisions. Nonslip walking surfaces and steps as well as grab irons for access to operating and servicing positions shall be provided and shall be located and spaced to permit safe dismounting on both sides of the crane. The access/egress system shall conform to SAE J185.

3.9.7 Electrical system. The crane shall be equipped with a 24-volt, negative ground, electrical system in accordance with SAE J821, J1127, and J1128.

3.9.7.1 Batteries. The batteries shall be size 6 TN in accordance with MS35000, and shall be readily accessible for servicing. The battery and cable terminals shall be identified with the respective polarily by raised or recessed markings and corrosion resistant nuts and bolts shall be provided.

3.9.7.1.1 Battery enclosure. A weather protected box or compartment-type battery enclosure shall be provided. The enclosure shall include a drain for

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water or electrolyte, a vent to prevent gas accumulation and shall provide accessibility for servicing without removal of the battery or other components. The enclosure shall be equipped with battery holddown components to prevent any damage to the batteries. The enclosure shall be internally treated to prevent corrosion.

3.9.7.1.2 Intervehicular connector. The crane shall be equipped with a 24-volt slave receptacle conforming to MS52131. The receptacle shall permit charging the crane-batteries and slave starting of the engine from an external source and charging and slave starting other equipment with the crane. The receptacle shall be installed on the exterior of the crane near the battery enclosure “and shall be accessible to personnel standing on the ground. The voltage potential as well as the slave receptacle shall be identified on a plate conforming to 3.18 which reads “slave receptacle, 24 volts" and is attached adjacent to the receptacle

3.9.7.2 Lighting system. A lighting system shall be furnished conforming to SAE J99 except that headlamps shall conform to SAE J598 and two taillights shall be furnished-with the second taillight being as far right as practical. Front and rest turn signals shall be furnished. Two floodlights universally adjust-able in both the vertical and horizontal plans a-rid conforming to SAE J598 shall be furnished with one mounted on each side of the boom and located such that they can be adjusted with the boom positioned at any elevation throughout its range.

3.9.7.2.1 Blackout lignting. One blackout headlight conforming, to MS51318 shall be mounted as close to the extreme left as practical, and positioned to provide the illumination with the minimum obstruction. Two blackout stoplight-taillights conforming to MS51330 shall be mounted adjacent to the taillights specified in 3.9.7.2 and shall be mounted in recesses or provided with guards. The blackout lights will be controlled by an interior switch which is readily accessible to the operator in his seated position and cannot be switched off accidentally. All other exterior lights, all interior lighting and the horn, if electric, will be made inoperable automatically when the blackout lights are switched on. A separate wiring harness may be provided for the blackout lights.

3.9.8 Brake system. A foot-controlled service brake system conforming go SAE J1152 shall be provided fog all wheels. The brakes shall be capable of bringing the crane to a complete safe stop within a 40-foot distance on a level paved surface from a travel speed of not less than 20 mph. A parking brake capable Of holding the crane on a 30 percent slope in-either direction shall be furnished. If a separate hydraulic system for the brakes is utilized, the brake system shall be filled with a fluid that is listed on the qualified product list of MIL-B-46176, and the contractor shall provide certification that the brake components utilized are compatible with brake fluid conforming MIL-B-46176. If air brakes are provided, air brake service and emergency line couplers conforming go SAE J318 shall be provided on the front of the carrier. A relay emergency valve shall also be provided.

3.9.9 Hydraulic system. The hydraulic system shall provide control of all hydraulic functions in all operations. Hydraulic filtration shall be in

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accordance with SAE J931. The pump or pumps shall power all crane functions with adequate speed and power. The system shall provide for simultaneous operation of swing and load hoist. Means shall be provided to prevent inadvertent dropping of the boom or load in the event of loss of hydraulic pressure or failure of a line. Means or techniques shall be provided to allow for retracting and lowering the boom and load and swinging the superstructure in the event of hydraulic power loss or line failure. Hydraulic hoses shall conform to MIL-H-52471. The hydraulic system shall be capable of operation on oil conforming to MIL-L-2104, grade 10 except that for low temperature operation; oil conforming to MIL-L-46167 may be used. The hydraulic system shall be capable of withstanding all operations without damage, deformation, leakage, or malfunction. The hydraulic system shall provide for fluid passage between the carrier and superstructure with continuous 360 degree superstructure " rotation.

3.10 Superstructure (crane). The revolving superstructure assembly shall be capable of continuous 360 degree operation.

3.10.1 Pedestal (boom support). The pedestal and base shall be mounted on the chassis to insure continuous 360 degree working range and shall withstand the forces caused by lifting, swinging, and traveling with rated loads. The pedestal shall provide a hydraulically operated boom hoist which will provide a maximum hook block height of not less than 30 feet from the ground at an operating radius of 10 feet with the crane level on outriggers on a level surface. If a counterweight is furnished, it shall be made of-a homogeneous material.

3.10.2 Boom. The boom shall be of the hydraulically operated telescoping type, consisting of the base section and one or more sliding sections to reach an operating radius of not less than 31 feet. Extension and retraction of the boom sections shall be accomplished by one or more hydraulic cylinders. Extension and retraction of the tip section may be accomplished by mechanical means provided that it can be accomplished by the operator in less than 15 . minutes. The boom tip shall be provided with not less than two guarded lubricated sheaves conforming to SAE J881 and a becket for two part line reeving. The crane shall incorporate the necessary mechanism that will prevent accident or damage from two-blocking.

3.10.3 Hook block. The crane shall be provided with a lubricated single sheave hook block with swivel and safety latch. The block shall be rated not less than 7-1/2 tons and shall be provided with a becket for ending the hoist cable at the hook block. The sheave shall conform to SAE J881. The hook block shall have adequate weight to insure payout of the wire rope without load on the hook when the boom and hook block are in any position.

3.10.4 Drum. The crane shall be Provided with a hydraulicallv driven hoist drum. The assembly shall include a self-locking mechanism or self-energizing brake capable of holding the maximum rated load. The drum pitch diameter shall be not less than recommended by SAE J881 for the wire rope furnished with the crane.

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3.10.5 Wire rope. The crane shall be provided with sufficient rope so that when the boom is fully extended at its maximum elevation, reeved for maximum part lines and the hook at ground elevations there shall be not less than three wraps on the hoist drum. Wire rope shall be in accordance with RR-W-410, type I, class 2, 6X19, construction 3 or 4, IWRC improved plow steel uncoated, preformed, regular lay. Wire rope shall have safety factors not less than recommended by SAE J959.

3.10.6. Boom angle and boom length indicators. A boom angle indicator shall be provided which shows angular elevation of the boom and conforms to SAE J375. The boom angle indication shall be visible to the operator in his normal seated position except when vision is blocked by the pedestal with the boom in position over the rear of the carrier. Means shall be provided for-the operator to determine the boom length in increments to match with the load capacity chart.

3.11 Electromagnetic emission. The electromagnetic radiation of the crane shall not exceed the recommended levels of SAE J551.

3.12 Toolbox. A permanently installed, readily accessible, weather protected, hinged, lockable toolbox or compartment shall be provided. It shall be of a size sufficient to contain all common handtools, special tools, and material not attached to other parts of the crane as required for reassembly after transporting (see 3.7). Drain holes shall be provided to prevent the accumulation of moisture.

3.12.1 Tools. All nonstandard or special tools required for operation and maintenance of the crane which are normally furnished to commercial customers and all tools required in the disassembly and reassembly of the crane for transport (see 3.1.1, 3.7.1, 3.7.2.1 and 3.7.2.2) shall be furnished.

3.13 “Lubrication. Adequate and accessible means of lubricating all surfaces requiring lubrication shall be provided. A lubrication chart identical to the chart provided in the technical manuals shall be furnished and mounted on each crane on a plate conforming to 3.18.

3.13.1 Lubricants. All parts and components shall be so designed that they are properly lubricated with specified military lubricants. The engine, transmission and torque converter shall operate on oil conforming to MIL-L-2104, except that for low temperature operation, oil conforming to MIL-L-46167 may be used. Initial lubrication and all subsequent lubrication shall be with the military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-M-7866 and MIL-G-10924. Unless otherwise preserved-fez packaging (see 5.1) all components shall be filled to the operating level with appropriate operational grades of military lubricants designated for use in the temperature range of the destination of the delivered crane. The crane shall be tagged in a conspicuous place to indicate the temperature ranges and grades of lubrication used.

3.13.2 Lubrication fittings. Fittings shall be in accordance with J534, shall be located in protected positions and shall be accessible to and compatible with a grease gun conforming to MIL-G-3859 equipped with a hydraulic coupler

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conforming to MS24207 and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts, except that engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed.

3.13.3 pressure-release device. A pressure-release device shall be provided where the use of pressurelubricating equipment will damage grease seals or other parts.

3.14 Fungus and moisture resistance.. All electrical components and connections, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.14.1 with 7 percent salicylanilide or 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish, except that:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating spall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the -varnish.

3.14.1 Composition. Composition shall be used unless prohibited by local air pollution regulations governing the application of the varnish. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.15 Lifting (suspension). tiedown, and extraction attachments.

3.15.1 Lifting attachments. The type I crane shall be provided with lifting attachments that comply with the requirements of MIL-STD-209, class 1 or 3, type II. The type II crane shall be provided with lifting attachments that comply with the requirements of MIL-STD-209, class 1 or 3, type II and MIL-STD-814. When the type II crane is sectionalized for helicopter transport (see 3.7.2.2), the attachments on each-section shall comply with MIL-STD-209, class 1 or 3, type II. Spreader bars are not permitted. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.15.2. The attachments shall withstand the forces in the amount and direction as specified without weld failure, damage, or permanent deformation of the crane or attachment.

3.15.2 Tiedown attachments. The type I crane shall be provided with tiedown attachments that comply with the requirements of MIL-STD-209, class 1 or .3, type II, -and MIL-A-8421. The type II crane shall be provided with tiedown attachments

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that comply with the requirements of MIL-STD-209, class 1 or 3 type II, and MIL-STD-814. Tiedown attachments may also be used as lifting attachments when such attachments meet the requirements specified in 3.15.1. The attachments shall withstand the forces in the amount and direction as specified without weld failure, damage or permanent deformation of the crane or attachments.

3.15.3 Extraction attachments. An extraction attachment located in accordance with figure 1 shall be provided on the type II crane to comply with the requirements specified in MIL-STD-814 for both low velocity airdrop and LAPES. The 'attachment shall withstand the forces in the amount and direction as specified without weld failure, damage, or permanent deformation of the crane or attachment. The pintle hook specified in 3.9 or an additional pintle may be used as the extraction attachment if it meets the requirements of this paragraph.

3.16 Identification marking, The crane shall be identified in accordance with MIL-STD-130. The marking shall be applied to the crane on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location. The plates shall furnish as a minimum, the contractor's model number, National Stock Number, USA Registration number, serial number, weight, 'shipping date and manufacturer.

3.17 Shipping data plate. Shipping data plate shall conform to MIL-P-514, type III, composition C, of type T, grade A, class 1 material and in addition, shall show the silhouette of the crane in transport position indicating the center of gravity and the location and capacity Of the lifting and the tiedown attachments. The plate shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.18 Instruction plates and decals. Each crane shall be equipped with instruction plates or decals, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the crane. Places shall be attached by screws, bolts, rivets or adhesive in a conspicuous protected location. The label lettering size shall be not less than 0.20-inches high and the size shall be graduated for- instructional or informational groupings with each group approximately 25 percent larger than each lower level group. The lettering width shall be approximately 3/5 the lettering heighth and the spacing shall be one stroke width between characters, one character width between words and 1/2 the character heighth between lines.

3.19 Vehicle marking. Lifting and tiedown attachments shall be identified with l-inch letters indicating "SLING POINT" Or "TIEDOWN". The tire pressure shall be marked on the frame near the wheels or wheel hubs with block or stencil-type letters not more than 1 inch and not less than 3/4-inch high. Stenciling and registration numbers shall conform to MIL-T-704.

3.20 Vehicle weight classification sign. A vehicle weight classification sign set (see 3.24) shall be applied to each crane. The crane vehicle classification data sheet shall be completed and submitted within the time frame specified (see 6.2). The contracting officer shall furnish the information required to apply the sign set".

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3.21 Decontamination unit mounting. A bracket assembly conforming to MS53052 shall be furnished. The bracket shall be mounted so that it is readily accessible to personnel with minimum contact with the crane and provide clearance to emplace and remove a decontaminating unit-conforming to MIL-D-51246.

3.22 Treatment and painting. All interior and exterior portions of the crane that are normally painted and all exposed surfaces that require protection from the environment shall be cleaned, treated, and painted in accordance with MIL-T-704, type F for ferrous metals and type G for nonferrous metals. The finish coat shall be forest green.

3.23 Servicing and adjustment. The contractor shall service and adjust all cranes for operational use including at least the following: Adjustment of lights; adjustment of the clutch and brake systems and mechanical components of the crane engine, transmission, and drive train; servicing of the cooling system with a 50-50 solution, by volume, of ethylene-glycol-type antifreeze conforming to MIL-A-46153, and water; and cleaning, bleeding, and filling of the hydraulic systems. The crane shall be completely lubricated as specified in 3.13.1.

3.24 Government-furnished equipment (GFE). The following will be furnished by the Government for use under the contract (see 6.5): .

a. Electrical Warning Decal Set, NSN 7690-00-267-5778.
b. Case, NSN 7520-00-55%9618 (the contractor shall furnish the fasteners and install the case and maintenance fores on the crane).
c. Vehicle weight classification sign set, NSN 9905-00-565-6207.

3.25 Workmanship. . .

3.25.1 Casting and forgings. All parts, components, and assemblies of the crane ‘which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework material shall meet typical mechanical properties of new tasted or forged material and not affect fit or function.

3.25.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to “ avoid overheating. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.25.3 Welding and welders. Welding procedures and welder qualification “ shall be in accordance with AWS D14.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contracts

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the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Component and material inspections. The contractor is responsible for insuring that components and materials used are manufactured, examined and bested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examinations.

4.3.1.1 Pretest examinations The type I and type II crane shall be subjected to the examination marked “X” under column 1 of table I. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post-test examinations. Upon successful completion of all tests specified in 4.3.2, the type I and type II cranes shall be subjected to the examinations marked "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3,1.1, the crane shall be subjected to the tests marked “X” under column 1 of table II. Failure of any test, any evidence of leakage, damage, breakage, or deformation after any or all of the tests or an unacceptable test report shall be cause for rejection.

4.3.3 Test report. The contractor shall Prepare and submit the first article test report to the contracting officer for approval within the time frame specified (see 6.2). This report shall include a detailed description of the conformance examination and demonstration performed, deficiencies encountered, and actions taken to correct the deficiencies.

4.3.3.1 Certifications. As part of the first article test report the contractor shall furnish the following certifications:

a. Unassisted starting and safe operation in any ambient temperature between minus 25° F and plus 120° F and at any altitude between sea level and 5000 feet.
b. Meeting requirements of OSHA, EPA, and PCSA Standard No. 2.

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c. If the crane is not equipped with a spark arrestor, certification from the engine manufacturer that the engine can be safely operated in ammunition, and petroleum oil, and lubricant supply areas.

4.4 Initial production inspection. When specified (see 3.3), one or more type I and type II initial production cranes will be selected at random by the Government from the cranes being produced by production tooling and will be examined for the defects marked "X" in column 3 of table I and subjected to the tests marked "X" in column 2 of table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of the intitial production crane(s) shall not exclude the remaining cranes from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the intial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the intital production test. Failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of the initial production crane to meet any requirements specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production cranes and shall be cause for refusal by the Government to continue acceptance of production cranes until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on cranes previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all cranes accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. After successful completion of all tests specified in 4.5.2, each crane shall be subjected to the examinations marked "X" in column 4 of table I. presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each crane shall be subjected to the tests marked "X" under column 3 of table II. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with table I.

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TABLE I. Examination schedule.

Preproduction		Initial Production	Quality Conformance	Defects	Requirement Paragraph(s)
Pre-Test	Post-Test				
1	2	3	4	5	6
X	-	X	X	101. Assembly not complete, components missing, or not as specified.	3.1
X	-	X	-	102. Weights and dimensions not as specified.	3.1.1
X	X	X	X	103. Material not as specified or rust, corrosion or deterioration has developed.	3.4
X	-	X	-	104. Keys and key slots not as specified.	3.4.1
X	X	X	-	105. Safety characteristic not as specified.	3.5.1
X	X	X	-	106. Human factors not as specified.	3.5.2
X	-	X	-	107. Sound level not as specified or warning plate not provided.	3.5.3
X	-	X	X	108. Back-up alarm not as specified.	3.5.4
X	-	X	X	109. Horn not as specified.	3.5.5
X	X	X	-	110. Maintainability not as specified.	3.6.1
X	X	X	-	111. Filling, draining or checking provisions not as specified.	3.6.2
X	-	X	X	112. Oil sampling valve not as specified.	3.6.3
X	X	X	-	113. Transportability not as specified or crane damaged from transport.	3.7

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TABLE I. Examination schedule. (Continued)

Preproduction Pre-Test	Preproduction Post-Test	Initial Production	Quality Conformance	Defects	Requirement Paragraph(s)
1	2	3	4	5	6
X	X	X	-	114. Reassembly kit not as specified (when furnished with type II crane only).	3.8
X	-	X	X	115. Carrier not complete, components missing or incorrectly installed.	3.9
-	X	X	X	116. Engine not as specified, does not function.	3.9.1
X	X	X	X	117. Fuel system leaks, damaged or not as specified.	3.9.1.1
-	X	X	X	118. Engine air induction system does not function and not as specified.	3.9.1.2
-	X	X	X	119. Charging system not as specified.	3.9.1.3
X	-	X	-	120. Engine housing not as specified.	3.9.1.4
X	-	X	-	121. Exhaust system damaged or not as specified.	3.9.1.5
X	-	X	-	122. Starter interlock system not as specified.	3.9.1.6
X	-	X	X	123. Tachometer drive not furnished or not as specified.	3.9.1.7
X	-	X	X	124. Transmission system not as specified.	3.9.2
X	X	X	X	125. Axles damaged or not as specified.	3.9.3
X	X	X	X	126. Steering system not as specified, damaged, leaks or does not operate.	3.9.3.1
X	X	X	X	127. Tires and rims not as specified or damaged.	3.9.3.2
X	-	X	X	128. Fenders not as specified or damaged.	3.9.3.3

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TABLE I. Examination schedule. (Continued)

Preproduction		Initial Production	Quality Conformance	Defects	Requirement Paragraph(s)
Pre-Test	Post-Test				
1	2	3	4	5	6
X	X	X	X	129. Outriggers not as specified, damaged, leaks or inoperative.	3.9.4
X	-	X	X	130. Operator's compartment not complete or as specified.	3.9.5
X	-	-	-	131. Instrument panel not complete or as specified.	3.9.5.1
X	-	X	X	132. Crane capacity chart missing, incomplete, incorrect or location not as specified.	3.9.5.2
X	-	X	X	133. Operator's cab not as specified.	3.9.5.3
X	-	X	-	134. Heater and defroster not as specified.	3.9.5.4
X	-	X	X	135. Boarding and servicing provisions not as specified.	3.9.6
X	X	X	X	136. Electrical system not complete, does not function or not as specified.	3.9.7
X	-	X	X	137. Batteries not as specified.	3.9.7.1
X	-	X	X	138. Battery enclosure not complete or as specified.	3.9.7.1.1
X	-	X	X	139. Intervehicular connector not as specified.	3.9.7.1.2
X	X	X	X	140. Lighting not as specified or does not function.	3.9.7.2
X	-	X	X	141. Blackout lighting not as specified.	3.9.7.2.1
X	X	X	X	142. Brake system not complete or as specified.	3.9.8

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TABLE I. Examination schedule. (Continued)

Preproduction Pre-Test	Preproduction Post-Test	Initial Production	Quality Conformance	Defects	Requirement Paragraph(s)
1	2	3	4	5	6
X	X	X	X	143. Hydraulic system not complete, leaks, damaged, does not function, or not as specified.	3.9.9
X	X	X	X	144. Superstructure not complete, missing components, does not function or not as specified.	3.10
X	X	X	X	145. Pedestal not complete or not as specified.	3.10.1
X	X	X	X	146. Boom not complete, damaged, inoperative, or not as specified.	3.10.2
X	X	X	X	147. Hook block not complete or as specified.	3.10.3
X	X	X	X	148. Hoist drum not complete, damaged, inoperative or as specified.	3.10.4
X	X	X	X	149. Wire rope damaged, incorrect size and length, or not as specified.	3.10.5
X	-	X	X	150. Boom angle and boom length indicators, damaged, does not operate or not as specified.	3.10.6
X	X	X	X	151. Tool box damaged, incorrect size or not as specified.	3.12
X	-	X	X	152. Tools missing or not as specified.	3.12.1
X	-	X	X	153. Lubrication means not provided as specified or chart not provided.	3.13
X	-	X	X	154. Lubricants not as specified.	3.13.1

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TABLE I. Examination schedule. (Continued)

Preproduction		Initial Production	Quality Conformance	Defects	Requirement Paragraph(s)
Pre-Test	Post-Test				
1	2	3	4	5	6
X	-	X	X	155. Lubrication fittings not as specified or not accessible.	3.13.2
X	-	X	X	156. Fungus and moisture resistance protection not provided or as specified.	3.14
X	-	X	X	157. Lifting, tiedown or extraction attachments damaged, missing, incorrectly located or not as specified.	3.15
X	-	X	X	158. Identification plate missing or not as specified.	3.16
X	-	X	X	159. Shipping data plate missing or not as specified.	3.17
X	-	X	X	160. Instruction plates and decals missing, not complete or not as specified.	3.18
X	-	X	X	161. Vehicle markings missing, incomplete or not as specified.	3.19
X	-	X	X	162. Weight classification sign missing or not as specified.	3.20
X	-	X	X	163. Decontamination unit mounting missing or not as specified.	3.21
X	-	X	X	164. Treatment and painting not as specified.	3.22
X	-	X	X	165. Crane not serviced and adjusted as specified.	3.23
X	X	X	X	166. Workmanship not as specified.	3.25

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4.6.2 Tests.

4.6.2.1 Test conditions. Prior to test, the crane shall be lubricated with oils and greases specified in 3.13.1 and serviced and adjusted as specified in 3.23. Oils shall be those designated for use in the ambient temperature at the place of test. Unless otherwise specified, tests shall be performed without shelter and at the climatic conditions existing at the time and place of test. All lights shall be turned on during all operational tests. Only that maintenance established by the contractor and submitted as scheduled maintenance prior to commencement of the tests shall be performed during the tests. Unless otherwise specified herein, he test area shall be examined to determine that:

a. There are no obstructions which may cause interference to raising the boom to its maximum height in any position or rotating the superstructure through 360 degrees.
b. Grade does not exceed 1 percent in any direction,
c. Test site is paved with hard surfacing.

prior to and throughout the testing the safety and health aspects of the crane shall be evaluated and reported on in the test report. The crane shall be checked prior to and during the tests to determine the following:

d. That coolant and lubrication systems are full.
e. Check that fuel tanks are maintained between full and half full level throughout these tests except for the road and craning tests.
f. Adjust clutches and brakes in accordance with the manufacturer's instruction manual. Inspection shall be made to insure that there is no dragging of the clutches or brakes with the controls in the disengaged position.
g. Examine all sheaves, revolving base rollers, drive chains, rotating gears, rotating drums and ropes to insure that there is no rubbing interference.

4.6.2.2 Test schedule. Tests shall be in accordance with table II.

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TABLE II. Test schedule.

Preproduction Test	Initial Production Test	Quality Conformance Test	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	X	X	Test conditions	4.6.2.1	3.13, 3.23
X	X	-	Maintainability	4.6.2.3	3.6
X	-	-	Crane lifting capacity	4.6.2.4	3.1.2, 6.8
X	X	-	Gradeability	4.6.2.5	3.1.2, 3.9.8
X	X	-	Speed	4.6.2.6	3.1.2
X	X	-	Road (3000 miles)	4.6.2.7	3.1.2
X	-	-	Lifting and rotating speed	4.6.2.8	3.1.2
X	-	-	Outrigger	4.6.2.9	3.9.4
X	X	-	Service and emergency brakes	4.6.2.10	3.9.8
X	-	-	Overall structural integrity	4.6.2.11	3.9.5.2, 6.8
X	X	-	Electromagnetic emission	4.6.2.12	3.11
X	X	-	Weights and dimensions and ramp angle	4.6.2.13(a)	3.1.1, 3.7.1
X	X	-	Rail	4.6.2.13(b)	3.7.1
X	-	-	Freefall (type II crane only)	4.6.2.13(c)	3.7.2.1
X	X	-	Airmobile (type II crane with reassembly kit only)	4.6.2.13(d)	3.7.2.2, 3.8
X	-	-	Turning diameter	4.6.2.14	3.1.2
X	X	-	Emergency steering (If equipped with full hydraulic steering)	4.6.2.15	3.9.3.1
X	X	-	Lifting attachments	4.6.2.16	3.15.1
X	X	-	Tiedown attachments	4.6.2.17	3.15.2
X	X	-	Extraction attachment (type II crane only)	4.6.2.18	3.15.3

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TABLE II. Test schedule. (Continued)

Preproduction Test	Initial Production Test	Quality Conformance Test	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	-	-	Boom	4.6.2.19	3.10.2, 3.9.9, 3.10.5
X	X	-	Craning	4.6.2.20	3.1.2
X	-	-	Sound level	4.6.2.21	3.5.3
X	X	-	Water fording	4.6.2.22	3.1.2
X	-	-	Low temperature	4.6.2.23	3.1.2
-	-	X	Production run	4.6.2.24	3.9, 3.10

4.6.2.3 Maintainability. Perform the maintenance procedures as outlined in the maintenance schedule submitted by the contractor prior to the test. Oil samples shall be taken from the engine and transmission. Inaccessibility to lubrication fittings, drainage of fluid on the crane, fluid leakage, inability to lubricate, incomplete listing in the maintenance schedule or inaccessibility or nonconformance to 3.6.3 of oil sampling valves shall constitute failure of this test.

4.6.2.4 Crane lifting capacity. With the crane stabilized on outriggers, lift, swing 360 degrees, and lower a load not less than 117.6 percent of the rated load at a 10 foot radius, 28 foot radius and at two additional intermediate radii. Repeat the above test without outriggers both through 36,0 degrees and over the front. Presence of a tipping condition, inability to lift, swing 360 degrees, lower the load or complete the test shall constitute failure of this test.

4.6.2.5 Gradeability. Without load and the boom in travel position, drive the crane up a 30 percent longitudinal slope of firm soil of not less than 30 feet in length. Repeat the above, stopping midway on the slope, apply parking brake, and hold crane on slope for not less than 90 seconds, release brake and continue up the slope. Repeat the above test descending the slope. Without a load on the hook, negotiate a 15 percent side slope of firm soil. Inability to ascend and descend the slope, inability to hold the crane on the slope with parking brake, inability to negotiate a 15 percent side slope, fluid leakage, damage or inability to complete the test shall constitute failure of this test.

4.6.2.6 Speed. Test the crane on a level, paved surface of sufficient length to attain maximum speed prior to entering the measured test course. Drive the crane through initial and final points that are not less than 1/4 mile apart and measure the elapsed time that is required to cover the measured distance.

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Repeat for a total of three times in each direction over the measured course to obtain the average clasped time in seconds. Average speed of less than 20 miles per hour shall constitute failure of this test.

4.6.2.7 Road (3,000-mile). Road test the crane for a distance of not less than 3,000 miles at an average speed of not less than 10 mph. One-half of this distance shall be over primary roads (see 6.8.5) at maximum sustained speeds, one-fourth over secondary roads (see 6.8.6), and one-fourth over unimproved roads (see 6.8.7) and off road (see 6.8.8). For a time period of not less than 10 hours the fuel consumption shall be measured for travel over paved highways at maximum attainable speeds Up to 20 mph. The unpaved course shall include a longitudinal slope of not less than 30 percent and not less than 30 feet of " uphill slope shall be negotiated every hour. The unpaved course shall also include 5 miles of rear-steer operation, of which 2 miles shall be rearward travel. Operating difficulties, insufficient fuel capacity for 200 miles at maximum attainable speeds up to 20 mph, noticeable lag between turning of steering wheel and steered wheels, or evidence of damage to any component shall constitute failure of this test.

4.6.2.8 Lifting and rotating speed. Lift a load equal to the maximum permissible line pull on a single hoist line and simultaneously rotate the superstructure. Repeat in the opposite direction. Without load and with the boom at a safe position, time with a stop watch 10 revolutions clockwise of the loaded supersturcture at governed engine speed. Repeat counterclockwise. Failure of the crane to lift the load and rotate the superstructure simultaneously or an average swing speed other than that specified in 3.1.2 shall constitute failure of this test.

4.6.2.9 Outrigger. Place 3 inch high wedge blocks under each outrigger pad in order to displace the edges and corners of the pads in the following arrangement:

a. Right rear outrigger raise the left front corner.
b. Left rear outrigger raise the right rear corner.
c. Right front outrigger - raise the right front corner.
d. Left front outrigger - raise the left rear corner.

Raise the crane to the maximum height on the outriggers, and rotate a rated load 360 degrees around the crane. Nonconformance to 3.9.4 or evidence of damage, breakage, or deformation shall constitute failure of this test.

4.6.2.10 Service and emergency brakes.. On a paved surface, accelerate the crane at maximum rate to a speed of not less than 20 mph. Apply the service brakes bringing the crane to a halt in highest gear. Immediately after the crane comes to a halts repeat for five additional cycles in the shortest possible time. From a speed of not less than 20 mph, apply the emergency brakes to bring the crane to a stop in the shortest distance. Measure the stopping distance of each stop by a paint or chalkmarking-type detonator instrument. Brake tests will be conducted with maximum pedal effort. Failure of the service brakes to bring the crane to a halt safely within 40 feet for each stop, failure-

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of the emergency brakes to stop the crane within the distance specified by SAE J1152, damage, failure or overheating of the brakes shall constitute failure of this test.

4.6.2.11 Overall structural integrity. The manufacturer shall submit a copy of the report of tests performed in accordance with SAE J1063. If tests have not been previously conducted on an identical crane, the completed first article shall be tested in accordance with SAE J1063. All resultant test data shall be furnished with the first article test report.

4.6.2.12 Electromagnetic emission. The crane shall be tested in accordance with SAE J551. Measured levels in excess of the recommended levels constitute failure of this test.

4.6.2.13 Transportability.

a. Weights, dimensions and ramp angle. Weigh and measure the crane for conformance to 3.1.1 and 3.7.1. Perform the ramp test in accordance with MIL-A-8421 with an upper barrier set at 102-inches above the upper horizontal ramp. Nonconformance to 3.1.1 and 3.7.1 or any interference between the crane and the ramps or the upper barrier shall constitute failure of this test.
b. Rail. Load the crane on a railroad flatcar in accordance with 3.7.1, and pass the railroad car through a mock-up of the PPI clearance diagram (figure 2). Inability to pass the crane through the mockup without interference shall constitute failure of this test.
c. Free-fall drop. This test shall be conducted by the Government at a Government facility after completion of all contractor conducted tests. The type II crane shall be rigged on the air delivery platform and drop-tested in accordance with MIL-STD-669. A minimum of two drop tests from a height of 12.7 feet shall be performed. Between drops, the crane shall be derigged, assembled, examined, started, all crane functions operated, and driven back and forth a few feet, to determine if any damage has occurred. The crane shall be sectionalized for airmobility in accordance with 3.7.2.2 and reassembled in accordance with the contractor's instructions to ascertain that no damage was done to the sectionalization features by the drop test. After the second drop, the crane shall be operated for not less than 8 hours, consisting of 4 hours of craning with various loads at the rated radii and 4 hours of loading on various road surfaces including off-road, to determine its ability to perform its intended functions. Failure to meet the rigging requirements of 3.7.2.1, weld failure, breakage or permanent deformation of any structural member or component, leakage, inability of the crane to perform its intended functions, inability to reassemble or reinstall any removed material or equipments inability to sectionalize and reassembly the crane shall constitute failure of this test.
d. Airmobile. Sectionalize the type II crane into two loads and mount the reassembly kit (see 3.8) on the loads. Weigh each load. With only the material and tools on the loads, assemble the loads into an operating crane. After the crane has been assembled and assembly kit

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stowed thereon, the crane shall be operated for not less than 2 hours to determine its ability to perform its intended functions. No other crane, hoist, tools, or equipment not carried on the two loads shall be used. Weight exceeding that specified in 3.7.2.2, inability to assemble in the time specified, by 'the specified personnel, use of material or equipment not a part of the load, or inability of the crane to function shall constitute failure of this test.

4.6.2.14 Turning diameter. Operate the crane in two complete turning circles in accordance with SAE J695 utilizing four wheel steer. A curb to curb turning diameter of more than 40 feet shall constitute failure of this test.

4.6.2.15 Emergency steering. If the crane is equipped with a full hydraulic steering system, the crane shall be required to negotiate the test courses defined in SAE J53 at 5 mph and 10 mph. Inability to negotiate the courses or excessive steering effort shall constitute failure of this test.

4.6.2.16 Lifting attachments. As applicable, the lifting attachments shall be tested to determine compliance with MIL-STD-109 and MIL-STD-814 based on the use of the attachment and the requirement specified therein. Weld failure, damage, or permanent deformation of the crane or the attachment shall constitute failure of this test.

4.6.2.17 Tiedown attachments. As applicable, the tiedown attachments shall be tested to determine compliance with MIL-STD-209 and MIL-STD-814, Weld failure, damage, or permanent defomation of the crane or the attachment shall constitute failure of this test.

4.6.2.18 Extraction attachments. On the type II crane, the extraction attachments shall be tested to determine compliance with MIL-STD-814. Weld failure, damage, Or permanent defamation of the crane or the attachment shall constitute failure of this test.

4.6.2.19 Boom. With the crane stabilized on outriggers and reeved for maximum parts of line, extend the boom hydraulically to its maximum length and measure the length from center of rotation to the centerline of the hook block. Retract the boom and elevate to the maximum angle, extend the boom to the maximum, and lower the hook block to the ground and determine the number of wraps on the drum. Inability to extend and retract, raise and lower the boom, less than three wraps on the drum, an operating radius less than 28 feet, any binding, leakage, damage, deformation or inability to complete the test shall constitute failure of this test.

4.6.2.20 Craning. Perform craning operations for not less than 100 hours, 50 hours before and 50 hours after the road test. Each 50 hour craning operation shall include 35 hours of lifting a load of not less than 10,000 pounds at the capacity chart rated radius, to a height of not less than 10 feet, swinging 450 degrees and lowering the load. The direction of rotation shall be reversed every hour. For 10 hours with the same load, lift the load approximately 1 foot off the ground and rotate the load continuously at not less than 1.75 rpm for periods of not less than 1 hour followed by rest periods of not more than 30 minutes. After each rest period, the direction of rotation shall be reversed.

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Carry a load as specified in 3.1.2(g) at an operating radius of not less than 20 feet over unimproved roads for not less than 5 hours at an average speed of not less than 1.5 mph. Inability to complete the, test, damage, deformation, loosening of components, binding, malfunction, leakage of fluid, or failure of any crane component shall constitute failure of this test.

4.6.2.21 Sound level.

4.6.2.21.1 Interior sound levels. The noise level at the operator's station shall be tested in accordance with SAE J919. The data shall include db (A) and octave band pressure levels taken at the operator's station. Duty cycle testing shall not be used.

4.6.2.21.2 Exterior sound levels. The crane shall be tested in accordance with SAE J88. The horn shall be operated and tested for conformance to 3.5.5.

4.6.2.22 Water fording. The crane shall be driven through a body of fresh water to a depth of not less than 30 inches ten times. On the tenth fording the crane shall be stopped at the 30 inch depth and the engine allowed to idle for not less than 15 minutes before -leaving the body of water. Immediately after the final fording, the brakes shall be checked for proper operation. All reservoirs and gear and bearing enclosures shall be sampled for presence of water prior to and after the test. Evidence of water leakage, damage, or malfunction of the brakes or any other components shall constitute failure of this test.

4.6.2.23 Low temperature. The crane shall be cold soaked to attain a stabilized temperature no warmer than -25° F. Temperature stabilization shall be measured at the following locations:

a. In the two center cells of each battery between the plates and at middepth of the electrolyte.
b. In the volumetric center of the hydraulic system reservoir.
c. In the engine lubricating oil gallery and in the oil sump.
d. In the engine coolant in the engine-cooling-jacket discharge pipe.

The crane shall be started at -25° F without preheating within 5 minutes of initial attempt and without the use of any external starting aids. Smooth engine running without unnatural or continued control manipulation shall be obtained within 15 minutes after starting. After the engine operation has stabilized and the hydraulic fluid has been allowed to reach proper operating temperature, all crane functions shall be operated, including lifting and rotating a rated load at the rated radius with and without outriggers. The heater and defroster shall be tested for conformance to 3.9.5.4. Temperatures will be taken at not less than 3 locations: at 2 inches above the floor between the operator's feet, at 24 inches above the lower seat cushion, and the operator's eye level. Repeat the entire test within a 48 hour period. Where a cold chamber is used to conduct this test and the crane cannot be rotated within the chamber, as much of the test shall be performed in the chamber as practical

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before removing the crane to complete the test. Failure to start at -25° F within 5 minutes, failure to obtain a smooth running engine within 15 minutes, nonconformance to 3.9.5,4, inability to operate or perform any crane function or evidence of damage, deformation, breakage or leakage of fluid shall constitute failure of this test.

4.6.2.24 Production-run. Each complete crane shall be tested in accordance with ANSI B30.15 paragraph 15-2.2.1 to insure correct assembly and adjustment. Inability to perform the operation, evidence of any malfunctions damage, deformation, breakage, leakage of fluid or inability to complete any test shall constitute failure of this test.

4.7 Inspection comparison. The Government may select cranes at any time during the contract production period and subject these cranes to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Goverment on units selected at random from those which have been accepted by the Goverment and will not include the previously inspected preproduction model and initial production cranes. In addition to any test specified as part of the inspection comparison, the Goverment reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison crane to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison crane(s) and shall be cause for refusal by the Government to continue acceptance of production cranes until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on cranes previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all cranes accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. The packaging shall be examined and tested in accordance with the quality assurance provisions of MIL-C-3580.

5. PACKAGING

5.1 Preservation and packing. The degree of preservation and packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A or B. Preservation, packing, and marking shall be in accordance with MIL-C-3580, except that antifreeze conforming to MIL-A-46153 shall be used.

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5.1.2 Level C. preservation and packing shall provide protection against deterioration-and damage and assure carrier acceptance and delivery to destination, without damage, at the lowest ratings of the carrier in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules. Gearboxes and crankcases shall be filled with operational lubricants and the cooling system shall be filled with a fresh clean solution of 50 percent water and 5O percent antifreeze conforming to MIL-A-46153. Marking shall reflect as a minimum, the item name, national stock number, quantity, contract number and appropriate address markings. The markings shall be applied with a waterproof material, be legible and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The cranes are intended for use in handling materials, lifting, and transporting loads short distances.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and" date of this specification.
b. Type of crane to be furnished (see 1.2 and .3.1).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. Number and type of preproduction model cranes, time frame required for submission, and whether the Government will conduct any or all of the examinations and tests. 'When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. When initial production testing is required and number of initial production cranes to be furnished (see 3.3).
f. When the reassembly kit shall be furnished (see 3.8).
g. When the heater and defroster is not required (see 3.9.5.4).
h. Time frame for submitting the vehicle classification data sheet (see 3.20).
i. Time frame for submitting test report (see 4.3.3).
j. Degree of preservation and packing required (see 5.1).

6.2.1 Proposal requirements. In the event that only type I cranes are required but there is a possibility of a later requirement for type II cranes, in order to insure that the requirement can be met through modification of the type I design the invitation for bids or the request for proposals should specify the requirement for adequate engineering analysis and information to show that the type I crane can be made to meet the type II requirements.

6.3 Preproduction model. Any changes or deviations of production cranes from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will. not relieve the contractor of his obligation to furnish cranes conforming to this specification.

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6.4 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification forms to the contractor and request the return of the completed forms to the Government. The information thus obtained should be utilized to determine the weight classification number(s) for the crane. The contractor should then be requested by the contracting officer to apply the weight classification number(s) to the crane utilizing the Government-furnished sign kit (see 3.20).

6.5 Government-furnished equipment. The contracting officer should arrange to furnish the Government-furnished equipment as required in 3.24.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each crane.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary -(including any special tools) and instructions on shipment of cranes. A suggested paragraph is as follows:

"Shipment of cranes shall include repair parts, maintenance tools, operational and maintenance instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.8 Definitions.

6.8.1 Rated load. A rated load is a load that is not more than 85 percent of the tipping load (see 6.8.4) in the least stable direction for a given radius with the crane resting on firm, level ground, with full cooling system, and with the fuel tank not less than one-half full. The structural strees levels specified by SAE J1063 shall not be exceeded.

6.8.2 Operating radius. The operating radius is the measured horizontal distance from the centerline of rotation of the revolving superstructure without load to the centerline of the crane hook block with the load suspended.

6.8.3 Tipping point. The tipping point is the condition of crane loading wherein the load moment acting to overturn the crane is equal to the maximum moment of the crane available to resist overturning.

6.8.4 Tipping load. The tipping load is the force acting to unbalance a crane at the tipping point (see 6.8.3). It includes the weight of the hook block and all items suspended from the hook block.

6.8.5 Primary roads. Primary road is defined as an asphalt or concrete road.

6.8.6 Secondary roads. Secondary road is defined as a well maintained dirt or gravel road.

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6.8.7 Unimproved roads. Unimproved road is defined as a primitive form of unmaintained road where the land was initially cleared and leveled by tractor or grader.

6.8.8 Off-road. off-road is defined as irregular terrain not previously leveled by vehicles.

6.9 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

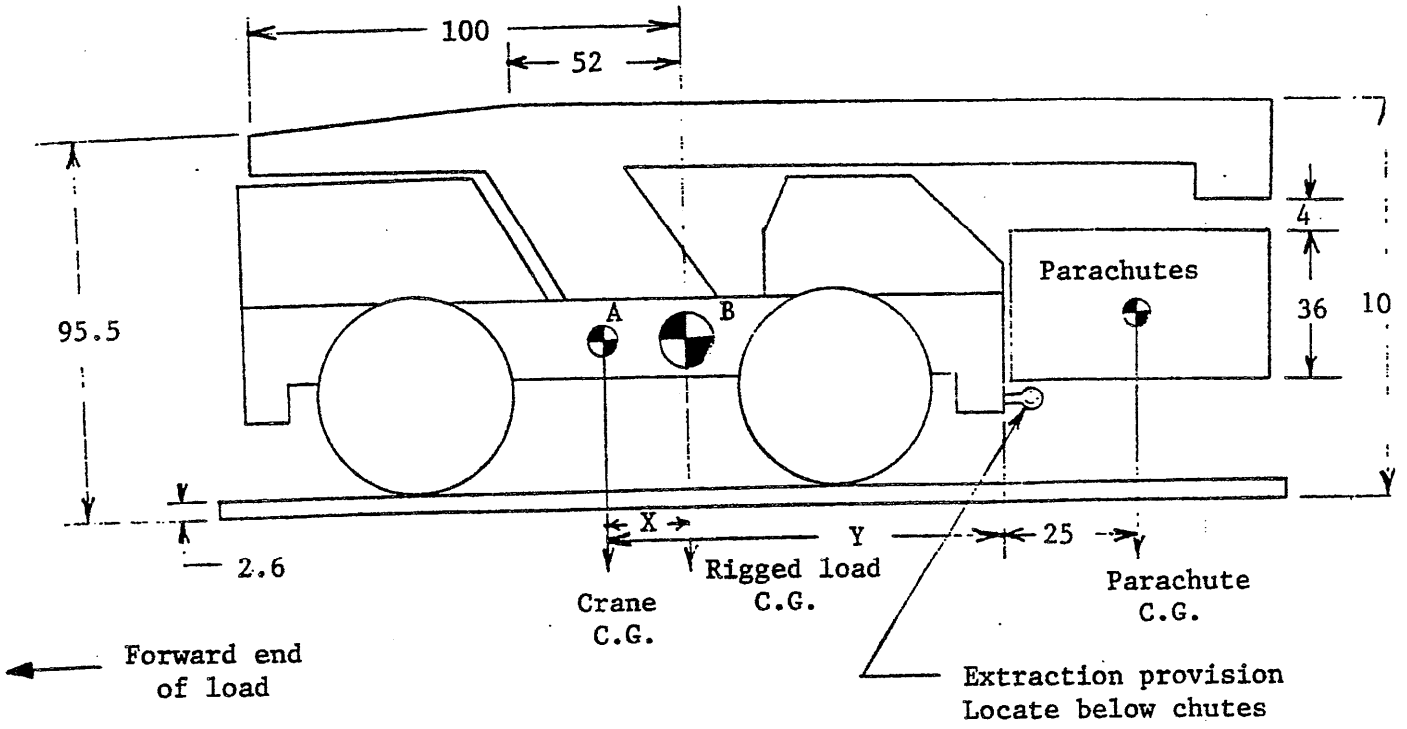
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Army - AT

Project 3810-A090

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NOTE: All dimensions are in inches.
The direction of the carrier may be reversed for improved center of gravity location and extraction provision installation provided that all requirement specified herein are met.

X = Distance from the C. G. of crane to the combined C. G. of rigged load.

Y = Distance from the C. G. of crane to the end used for extraction.

A = Weight of the unrigged crane.

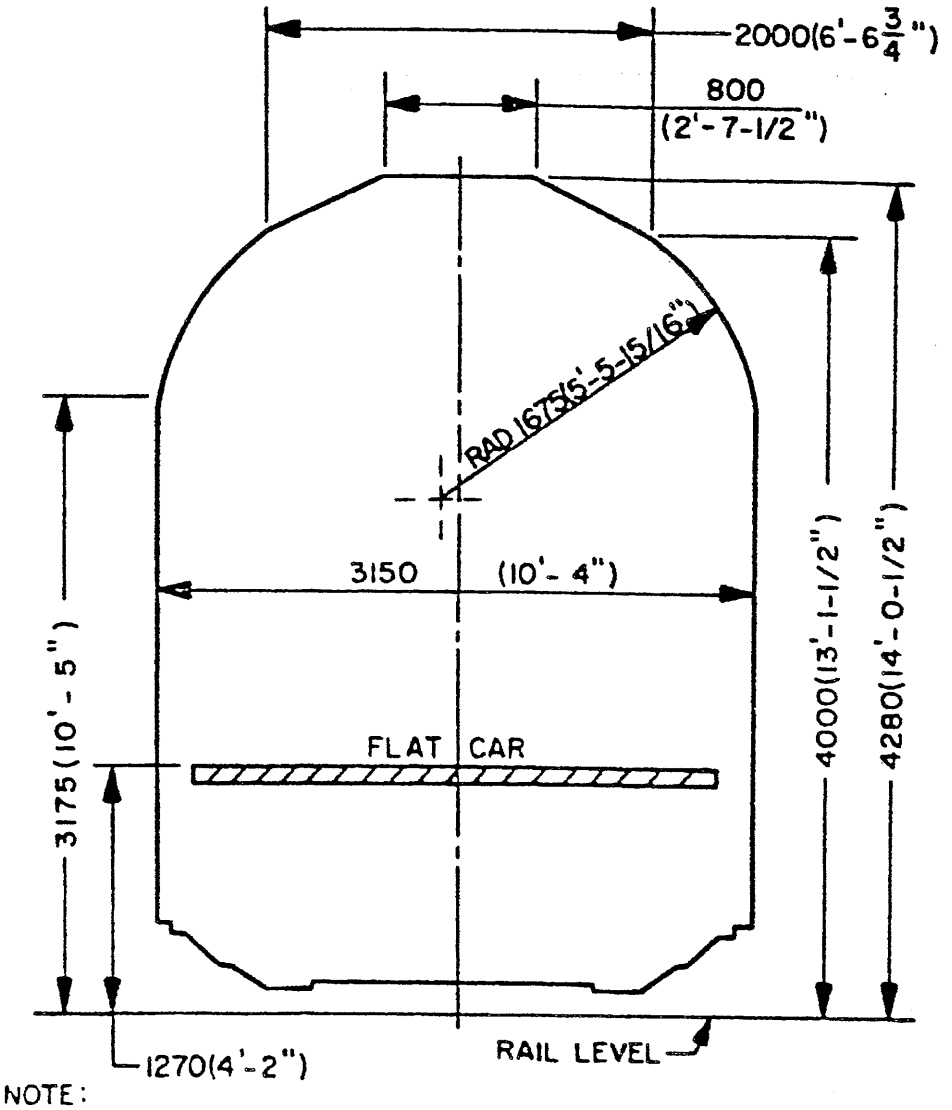
B = Weight of the rigged load = A + 4,300 lbs.

$$X = \frac{2300\ (Y+25)}{B}$$

FIGURE 1. Determination of center of gravity location for rigged load.

X-4119

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NOTE:
DIMENSIONS SHOWN THUS: MILLIMETRES(FEET-INCHES)

FIGURE 2. Passe-partout international (PPI) gauge.

X-4093A

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NSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in ubmitting comments and suggestions for improvements. All users of military standardization documents are invited to provide uggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and nailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was oo rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the roblems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an cknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being onsidered.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52974(ME)	Crane, Wheel Mounted, Hydraulic, Light, 7-1/2 Ton

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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